

Patrick Norton

Reed College MS 1099, 3203 SE Woodstock Blvd, Portland, OR 97202-8138

✉ pnorton@reed.edu

☎ +1 734 263 0562

🌐 github.com/PatrickNorton

Education

University of Toronto

PhD Electrical & Computer Engineering (upcoming)

Toronto, ON

September 2026–

Reed College

BA Math/Computer Science

Portland, OR

Aug 2021–May 2025

GPA: 3.78

Relevant Courses Taken

- Computability & Complexity
- Algorithms
- Abstract Algebra
- Lie Algebras
- Deep Learning
- Linear Algebra (proof-based)
- Number Theory
- Computer Systems
- Topology
- Vector Calculus

Research Experience

Vector Institute and the University of Toronto

Internship

Toronto, ON

Sep 2025–Sep 2026

- Research on validating fairness of machine-learning models.
- Used zero-knowledge proofs to be able to validate fairness without leaking public information.
- **Learning outcomes:** Computer-science research skills, mathematics of machine learning, applying zero-knowledge proofs.

Reed College

Senior Thesis

Portland, OR

Aug 2024–May 2025

- Expository thesis on probabilistically-checkable proofs and zero-knowledge proofs.
- Presented results from multiple recent papers in cryptography and complexity theory.
- **Learning outcomes:** Presenting mathematics in an accessible way, probabilistically-checkable proofs, zero-knowledge proofs, algebrization.

Reed College

Grader, Math 332

Portland, OR

January 2024–May 2024, January 2025–June 2025

- Graded Math 332, Reed's abstract algebra course.
- **Learning outcomes:** Reading mathematical writing, abstract algebra skills.

Reed College

Peer tutor, Math

Portland, OR

August 2024–December 2024

- Ran peer tutoring for students in Reed's first and second-year math courses.
- **Learning outcomes:** Tutoring, explaining mathematics, working with students.

Reed College

Internship

Portland, OR

May 2024–August 2024

- Worked with Prof. Zaji Daugherty on decomposing Lie algebras into highest-weight spaces.

- Generated decompositions for the symmetric algebra.
- Presented research results at a poster session.
- **Learning outcomes:** Knowledge of Lie algebras, learning about the process of math research, combinatorial algebra skills.

Reed College

Grader, CS 221

Portland, OR

August 2022–December 2023

- Graded CS 221, Reed's second-year computer science course, focusing on C++ and assembly.
- **Learning outcomes:** Debugging code, learning floating-point internals.

Los Alamos National Laboratory

Internship

Los Alamos, NM

May 2022–August 2022, May 2023–August 2023

- Worked on DIORAMA, a nuclear-and-satellite simulation program used by the US government.
- Helped modernize code and remove technical debt, as well as refactoring the testing framework.
- Created a library for femtosecond-precision timekeeping (*femtotime*), able to work with both UTC and GPS time as well as converting between the two.
- **Learning outcomes:** Debugging and modernizing production C++, familiarization with debugging systems, learning about UTC and GPS time, learning about different location schemes.

Dartmouth University

Internship

Hanover, NH (remote)

May 2021–Aug 2021

- Assisted Prof. Sean Smith's graduate students with creating a debugger for a parser.
- **Learning outcomes:** Introduction to C, knowledge of parser systems.

Greenhills School

Senior Project

Ann Arbor, MI

Jan 2021–Apr 2021

- Implemented a lock-free concurrent-vector in Rust.
- **Learning outcomes:** Introduction to concurrent programming and algorithms, familiarization with concurrent debugging techniques.

Publications

- Mark Galassi and **Patrick Norton**. *femtotime*. Computer Software. USDOE National Nuclear Security Administration (NNSA), May 2024. DOI: 10.11578/dc.20240517.1. URL: <https://github.com/lanl/femtotime>

Other Experience

Math Corps at U(M)

Community Assistant

Ann Arbor, MI

July 2025–August 2025

- Assisted teaching mathematics to middle and high schoolers.
- Helped grade homework and come up with problem sets.
- Helped foster a positive community among campers.

Way-of-Life Martial Arts

Assistant Instructor

Hamburg, MI

September 2019–August 2021

- Assisted teaching karate classes to young children.

Blue Lake International

Blue Lake, MI

Musician

July 2018 & July 2019

- Played trombone in an international concert band.
- Toured in Germany, France, the Netherlands, and Poland.

Relevant Skills & Languages

- Programming Languages
 - Fluent: Rust, Java, Lisp, C++
 - Proficient: C, Python, Haskell, MIPS assembly
- Linux, MacOS, Emacs, L^AT_EX

Honors & Awards

Florence H. Leslie Math Prize		Reed College	
	June 2021	Academic commendation	2021–25
National Latin Exam		Phi Beta Kappa	
Summa cum Laude	2019		2025